

Anatomy Trains in motion

Thomas Myers

Introduction

Given the new understanding of what Schleip has termed the “neuro-myo-fascial web” (Schleip, 2003) (Chapter 1), let us now turn our attention to functional chains of myofascia, known as “myofascial meridians” or the Anatomy Trains, to explore some implications of this point of view for movement training (Myers 2001, 2009, 2013, 2020).

Muscle limitations

For the last 400 years, our guiding light in understanding movement has been the concept of “a muscle” (Vesalius, 1548). Movement has been understood to be an interplay between the forces in the surrounding universe—gravity, inertia, friction, momentum—and the force generated by the ~600 named muscles working in concentric, eccentric or static contraction across joints limited by bone shape and ligamentous restriction (Hamilton et al., 2011). Muscle actions have been principally defined in terms of their origin to insertion on bony attachments (Muscolino, 2002).

Thinking more systemically, in the light of recent research (Barker et al., 2004; Vleeming and Stoeckart, 2007), the concept of a muscle working only to draw together its ends appears to be a limited metaphor: useful for getting here but now so outdated that it requires several caveats. For example, it is now clear that muscles also attach to other muscles along their sides

(Huijing, 2007). The far-reaching implications to force transmission and muscle mechanics are just being explored (Maas, 2009). The elasticity of large connective structures, such as tendons, changes our thinking about force transmission and efficient movement (Kawakami et al., 2002). Muscles also attach to and affect nearby ligaments (Van der Wal, 2009). The epimysium also attaches to nerves and neurovascular bundles which serve that muscular tissue (Shacklock, 2005).

Thus, the origin-to-insertion standard theory of muscle action misses out on (at least) the following four aspects, which we now know need active consideration:

1. Force transmission via intermuscular fascia to nearby muscles.
2. The ability of muscle tension to reinforce nearby ligaments.
3. The pull on nearby neurovascular bundles.
4. Force transmission from segment to segment via fascial continuities spanning the joints (Tietze, 1921; Franklyn-Miller et al., 2009).

Fascial neurology

On the neurological side, the nervous system appears to be about six times more sensorially interested in what goes on in the fascial matrix than it does in detecting changes in the muscle itself (Van der Wal, 2009; Grunwald, 2017).

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In other words for each muscle spindle detecting length changes, there is an order of magnitude more stretch receptors in the fascia that, depending on their arrangement, can detect load, stretch, vibration, pressure, shear, or convey pain.

Furthermore, no representation of individual muscles has been found within the sensory or motor cortex of the brain. The nervous system works in terms of a cybernetic, or self-regulating system, managing individual neuro-motor units governing in the order of 10–100 muscle cells within a muscle (Williams, 1995). The brain organizes movement in terms of coordinating these individual units, but not as our anatomy books do, in terms of individually named muscles. This means that the topology of contraction and fascial plane movement within muscles (perimysium) and between them (intermuscular septa) during any motion are far more complex than we have previously assumed (Fukunaga et al., 2010).

Biotensegrity

Another limitation to our current thinking is indicated by the “biotensegrity” model, which has been applied to the whole body (Fuller, 1975; Levin, 2003; Myers, 2009; Scarr, 2008), and to cellular structure and peri-cellular mechanotransduction (Ingber, 1998; Horowitz, 1999). This engineering model allows us to go beyond the usual Newtonian thinking in terms of the forces and vectors of muscles on joints, and enables us to see how our 70 trillion cells hang together in one organism as an “adhesome” (Zaidel-Bar et al., 2007; Zamir and Geiger, 2001), and how the “neuromyofascial web” accommodates movement as a self-adjusting whole. The Anatomy Trains maps out

a set of connected tissues in the parietal myofascia that essentially form the outer tensional network, which pulls in on the skeleton to help keep it erect and in the proper relationship (or not, in dysfunction).

In summary, from either a fascial, neurological or biomechanical point of view, the ubiquitous concept of “a distinct muscle” turns out to be an artifact of our common dissection method, and is neither a biomechanical nor a neurological reality. This idea has yet to penetrate the professional populace working with the public in either rehabilitative or training terms.

Once we reject the old idea, the significance of all these new findings taken together can be summed up in one word: resilience. All the factors named above contribute to the resilience of human tissue and to rapid global and local distribution of strain and differentiated response of the organism as a whole. The “isolated muscle theory” that has predominated our thinking has limited our perception of this body-wide “give” that is essential to resilience.

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The Anatomy Trains myofascial meridians map is one small aspect of this larger vision, and concerns consistent longitudinal connections within the singular fascial webbing. It posits myofascial force transmission, at the very least in the stabilizing of movement and in postural compensation, from one myofascial unit to another along these lines. To construct the Anatomy Trains map, we look for consistent fiber direction and fascial plane level. In this light, 12 myofascial meridians, each of at least three muscles, have been described.

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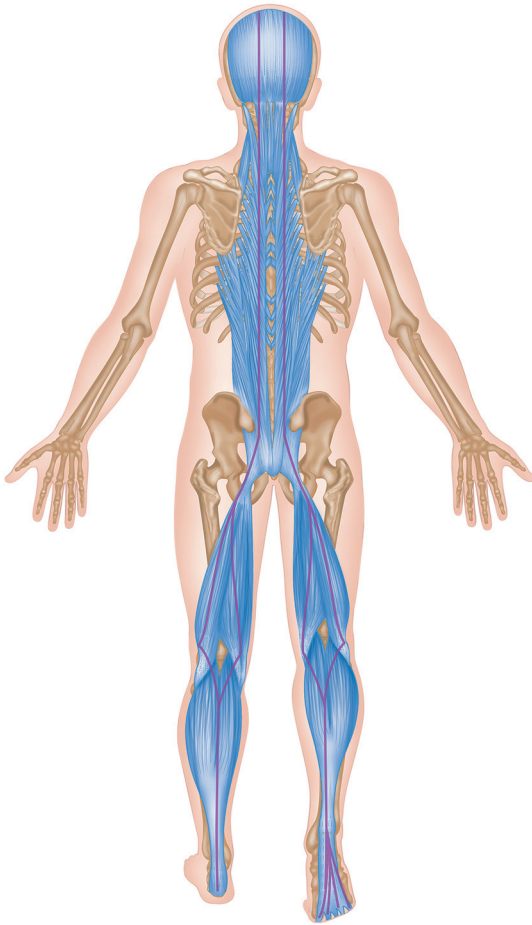


Figure 36.1 Superficial back line

The superficial back line, which traverses the posterior aspect of the body from toes to nose, operates functionally to bring our eyes up to positions that satisfy our curiosity, and lifts our body upright and keeps it posturally stable.

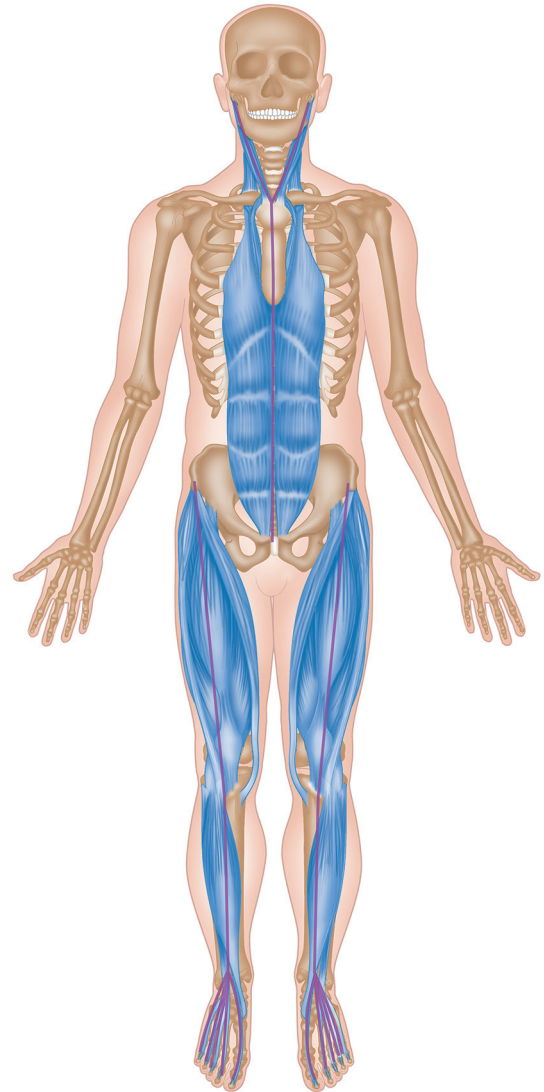


Figure 36.2 Superficial front line

The superficial front line, which traverses the anterior surface of the body, protects the ventral cavity and is thus associated with the startle response, and creates trunk flexion with leg extension.

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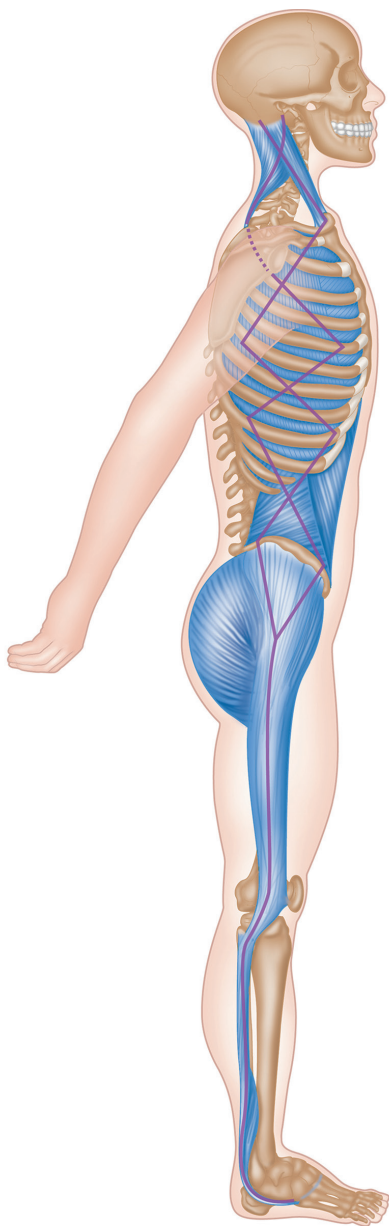


Figure 36.3 Lateral line

The lateral line, which runs from outer arch to ear, operates to create lateral bends or to prevent lateral bends to the opposite side. Thus, the lateral line operates to maintain stability during locomotion.

Trikonasana

Although difficult on the printed page, let us put the anatomy trains map into motion, by looking at a common but complex yoga pose, Trikonasana (or triangle pose), in terms of the Anatomy Trains lines (see Figure 36.8). A good analysis would require seeing the pose executed to both right and left, to see the differences from right to left. Based on this one photograph, we can see good form in the legs, with the lateral line on the left side being evenly elongated from the outer ankle to the hip, and the deep front line up the inside back of the right leg up to the pelvic floor on that side.

In the upper body, we see more compensation, not in the upper lateral line, but in the upper spiral line. From her right anterior superior iliac spine (ASIS), the internal and contralateral external oblique and the serratus anterior on the left ribcage is sufficiently short to pull the left ribcage forward so that the breastbone faces down towards the ground. The extra twist is required as compensation in the neck and in resisting the left shoulder coming forward.

Conversely, the left spiral line, from the left ASIS passing down and under her right ribcage to the right scapula and on to the left side of the head, could be considered to be too weak and in need of some strengthening to bring the right ribs further forward and allow the twist in the torso to straighten. We can wager that, if it is an imbalance in the spiral line, the pose would look significantly different on the other side.

Perhaps our premise is wrong and the inability to rotate the torso is not due to an imbalance in the spiral line but is in the rotational muscles closer to the spine, perhaps the psoas complex in the front or the multifidus complex in the back. We would test this with palpatory or movement assessments to isolate whether the restriction was in either the neuro-motor patterning or the fascia connected to the spiral line, superficial back line, or the deep front line. Any or all of these together could be the cause of this apparent limitation in movement. Of course, it is possible

(Continued)

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that the compensation is due to a structural anomaly within the spine, which we could detect by “end-feel” in the movement assessment.

The author makes no claim as to the exclusive nature of the Anatomy Trains. Firstly, those with significant deviance from the usual structure, as in a significant scoliosis or alteration from trauma, may create their own fascial “lines of transmission”. Secondly, the actuality of force transmission along the myofascial meridians is strongly suggested by some literature (Vleeming and Stoeckart, 2007; Franklin-Miller et al., 2009) but has yet to be proven by scientific research. The author is confident that something of this nature will prove to be the case, but meantime *caveat lector* (reader beware).

Proportional movement

In the stability/mobility (stiffness/control) modeling of human movement, the anatomy trains contribute to both. The key to the difference can be summed up as “proportional movement”.

All biological structures are granted a bit of “give”. Even living bone demonstrates resilience, although it reduces as the body grows older. Every softer tissue, from cartilage to ligament to tendon to fascia to nerve to every other named tissue, will bend or stretch or deform either a little or a lot before it tears or breaks.

In movement, tissues are commonly taken beyond their resting length into a process of stretch. The stretch is translated into both the cells, of all tissues along the line, as well as the

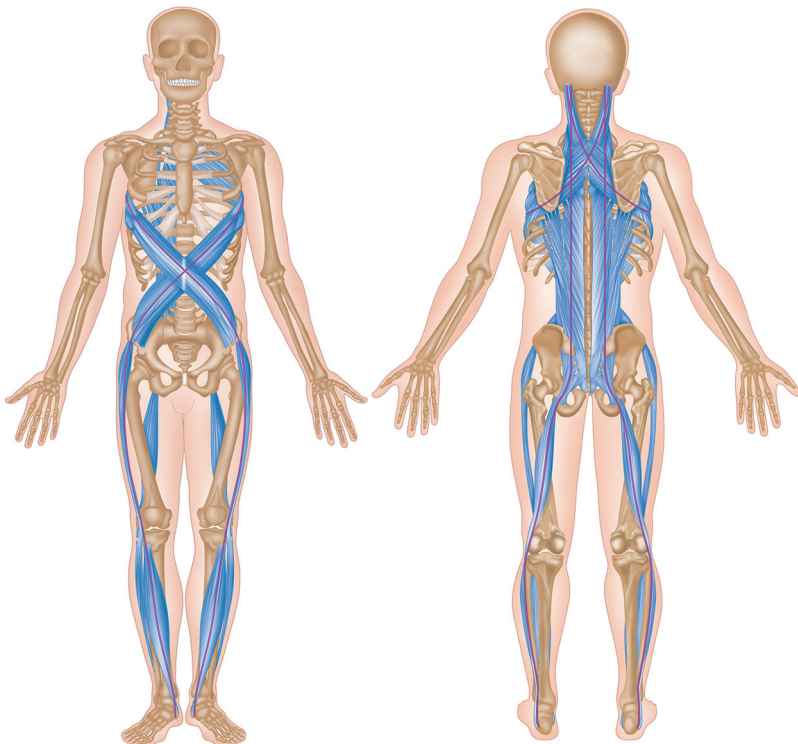


Figure 36.4 Spiral line (front and back view)

The spiral line winds around the body through the previous three cardinal lines, creating and modulating rotational and oblique movements in gait and sport.

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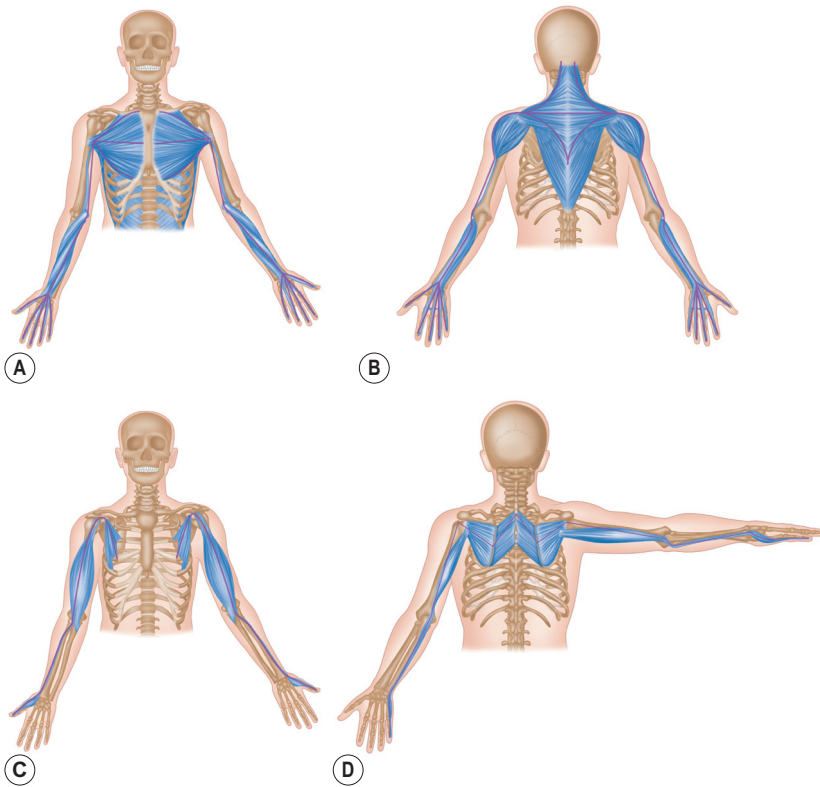


Figure 36.5 Arm lines

The four arm lines stabilize and move the arms and shoulders through their wide range of motion.

fibers and gels of the interstitia (Ingber, 2006; Langevin et al., 2006).

Given that the bearing point between the bones, at any given moment, constitutes the fulcrum for such tissue stretch, and given that the exogenous forces bearing on the tissues are within that tissue's ability to stay intact, then we can logically conclude that the largest movement will be near the skin, farthest away from that axial point. The rim of the wheel moves more than the hub. Therefore, the sensory endings in the skin will be most sensitive to initiated movement.

Also, if we accept the notion that the body will distribute such strain both laterally and longitudinally, then there must be a slight “give” or resilience in the tissues around the joints.

These tissues are called “muscle attachments” in classical theory, or more generally “peri-articular tissues”. In the Anatomy Trains metaphor we call them “stations” to indicate that, even although the connective tissues in those areas are stuck down to the underlying joint capsule or periosteum, there is a fundamental continuity of the connective tissue fibers in both direction and plane into the next segment and often onto a different muscle.

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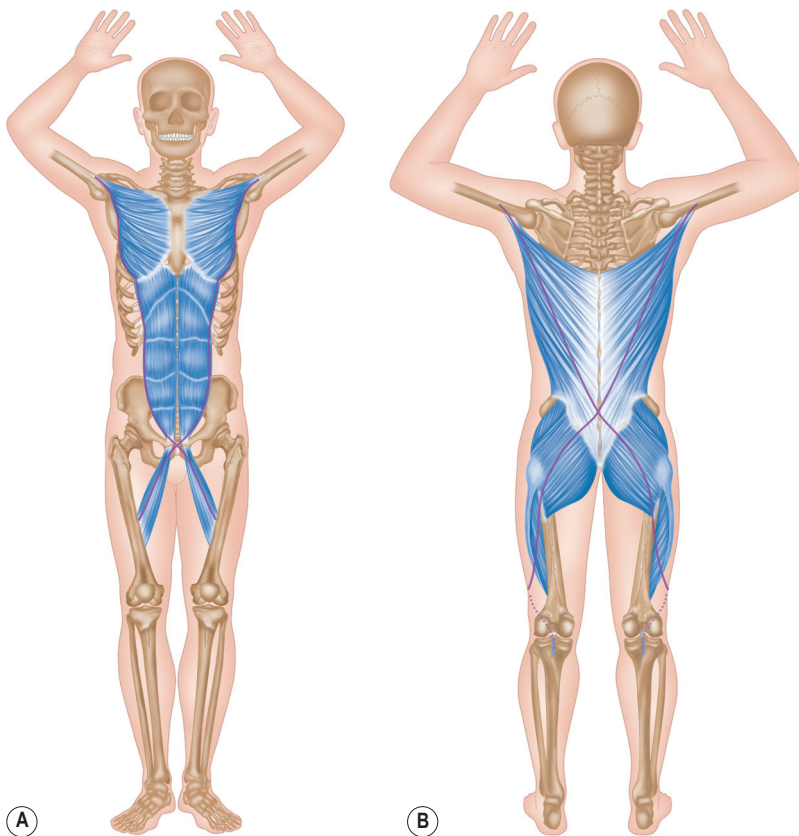


Figure 36.6 Functional lines

The three functional lines stabilize the shoulders to the contralateral and ipsilateral legs, extending the lever arms of the limbs through the trunk.

Resilience

As an example of resilience, the spiral line posits a continuity from the tibialis anterior of the anterior crural compartment with the anterior portion of the iliotibial tract, which in turn connects to the tensor fasciae latae and thus to the anterior superior iliac spine.

The lateral line suggests a connection from the long fibularis muscle (peroneus longus) across the fibular head at the superior end of the lateral compartment onto the central part

of the iliotibial tract and up via the aponeurosis of gluteus medius to a wide attachment on the iliac crest. Deep to this line would be the lateral collateral ligament (LCL), connecting the fibular head to the lateral condyle of the femur just above the joint capsule. Deep to the LCL would be the joint capsule itself, which includes the cruciate ligaments.

Although these fascial structures are identified separately, the outside of the knee is a fascial continuity from skin to bone, so all of these structures are connected. There is no discontinuity in

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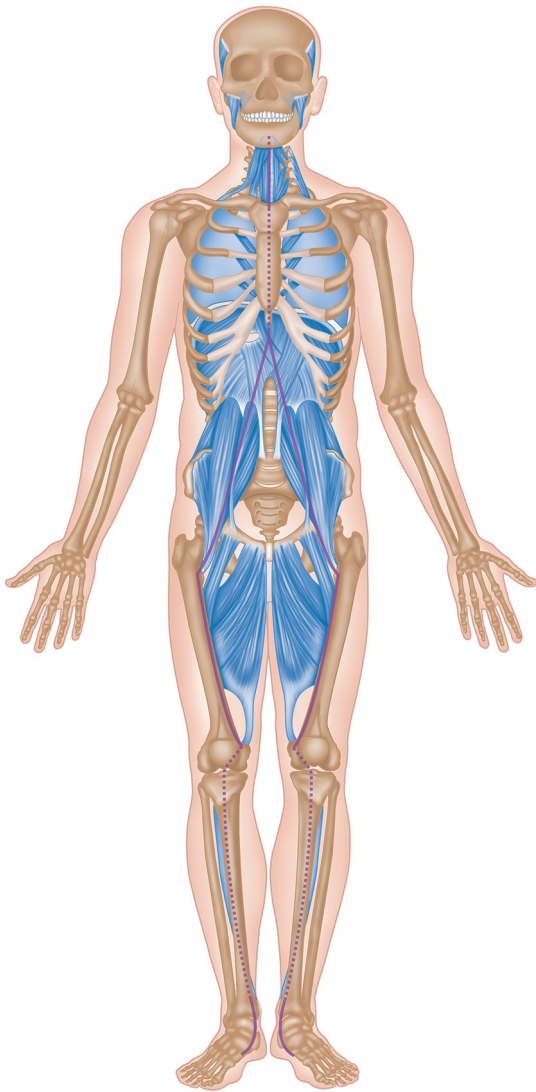


Figure 36.7 Deep front line

The deep front line, which runs from the inner arch to the underside of the skull and includes all of what could be termed the body's core, supports the stability and axial-appendicular extension in all our movements.



Figure 36.8 Analysis of a yoga pose

Putting our hands just below the bra line on either side and encouraging the ribcage toward a left rotation will reveal which tissues are restraining the motion, so that specific and appropriate movement cues or manual therapy may be applied to create even tone along the line.

the tissue (Guimberteau, 2004). So the following question arises: In day-to-day functional movement, or in more strenuous high-performance movement occasioned by sport, dance, or an impending injury, can the tissue “give” enough to allow the force to be distributed more widely or will all the force be focused on a particular structure, which is then more likely to fail?

Obviously, too much “give” in these structures would contribute to joint instability and

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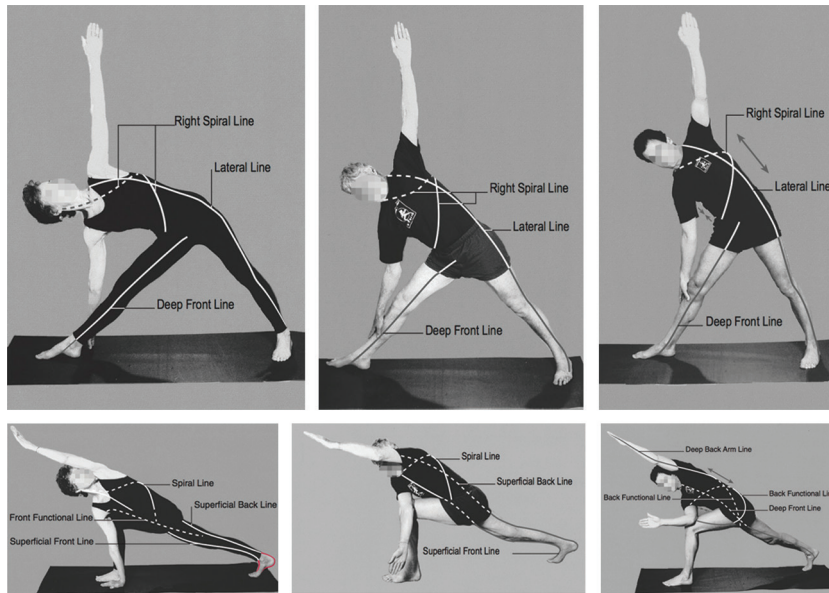


Figure 36.9 Yoga photographs from anatomy trains

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In the pictures, we can see a teacher, an experienced student, and a neophyte, all performing the same poses. We can readily see how the lines become straighter and more even in tone with developing practice. This palintonicity is a hall-mark of elongated and balanced posture, stabilized functional movement, and provides the greatest access to resilience along and deep to the lines.

those with “ligamentous laxity” can experience joint subluxation or malalignment fairly easily (Milhorat et al., 2007). Less obviously, too much stiffness means more strain localization, which can contribute to local structures, the LCL, medial collateral ligament or anterior cruciate ligament, for instance, being strained to the point of tearing.

The ideal lies somewhere in the middle: enough give to allow for strain distribution but not enough to allow for too much joint play. The factors involved in the ideal set point depend on the genetic predisposition to fascial build-up within the individual, as well as their training and performance demands.

If we could magically observe, in a healthy living body, a cross-layer slice from the skin down to the outside of the knee capsule during a long lateral reach, like the top of a tennis serve, we would see each successive layer moving on the one below from the surface down to the nearly immovable bone-to-bone layer. This resilience, or lack thereof, can be felt when placing the hands inside and outside a knee when such a movement is performed. By doing this, with a number of different subjects, it is easy to tell where the resilience is too tough, so non-compliant as to lead to a tearing injury in some nearby weaker tissue, or too tender and overly compliant, thus likely to lead to joint displacement.

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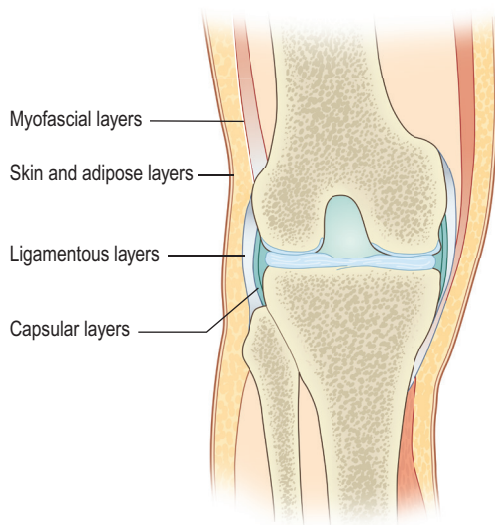


Figure 36.10

Tissue resilience.

In the case of too tough, specified sustained precise stretching or manual work is called for to increase the resilience across these anatomy trains stations that link the muscles across the joints. Cases that show too compliant tissues would require increased muscle tone in the myofascial tracks that support the stations, in this case the extensions of the abductors and the fibular muscles. This serves simply as an example of a process going on throughout the body. We could imagine a vertical view from the gluteus maximus to the deepest part of the sacrotuberous ligament in a forward bend or the ubiquitous yoga asana, Downward Dog. We would expect:

- the gluteus to give way easily into an elastic stretch
- the superficial part of the sacrotuberous ligament to link and distribute strain from the hamstrings up to the ‘myofascialature’ of the thoracolumbar area

- the deep portion to maintain the necessary relationship between the sacrum and ischium, allowing only a little give in the sacroiliac joint for the couple of degrees of movement required.

Excess chronic muscle tension or fascial adhesion anywhere in this linked chain could skew the distribution of these forces, creating the conditions for injury, postural compensation, known to trainers as “bad form”, or chronic overuse of tissues, leading to inflammation and pain.

As another example: in a side bend, can you see or palpate the progressive movement from the lateral abdominal obliques through the lateral edge of the iliocostalis lumborum, and the lateral raphé of the abdominals through the quadratus lumborum to the intertransversarii, to the intertransverse ligaments near the spine? Lack of resilience in any of these structures will create functional aberrations. The skilled practitioner will apply manual or movement therapy to the precise structures needing either more resilience or more tone.

Give a little

Small movements in the profound tissues allow the big movements closer to the surface. The movement of the dural membranes commonly palpated in cranial osteopathy (Sutherland, 1990) or the mobility of the organs assessed in visceral manipulation (Barrall and Mercier, 1988) are other common examples of small inner movements that have a significant effect on the large outer movements if they are missing or aberrated.

In the parietal myofascia, which is the domain of the Anatomy Trains, most trainers, physiotherapists, and manual therapists observe this effect in the tiny adjustments in the sacroiliac joints, which have a wider impact on the larger

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movement of the body as a whole. With too little sacroiliac movement, there is a large compensatory pattern in the gait and thus a tendency to “export” the problems to the lumbar spine. With laxity and too much sacroiliac movement, the outer body strives to compensate for too much inner movement by using the outer myofascialature as straps.

Across the entire body, from the spine to the limbs and across all the anatomy trains lines, we are looking for that “Goldilocks” feeling of being “just right”. This varies between subjects depending on natural fascial tone. It requires on-the-job training in accurate palpation of the anatomy trains stations in the relative movement of inner structures to set the stage for the proper movement of the outer structures.

Returning to our knee example, soccer players require stability generation on the inside and

outside of the postural leg when planting it in preparation for kicking a football with the opposite foot. As that foot lands, and the player seeks stability, differences in the turf or the player’s angle require near instantaneous adjustability in the foot, ankle, knee and hip.

As a test for this adjustability and resilience through the knee, have the player stand in front of you as you kneel and cup your hands around the inside and outside of the knee, covering the lateral line tissues on the outside, and the deep front line tissues (from the pes anserinus down to the medial collateral ligament) on the inside. Have your client, while using the other foot or a handhold for balance and keeping the planted foot on the ground, slowly translate their pelvis laterally from left to right. This will naturally create abduction and adduction at the hip, and inversion and eversion at the subtalar joint at the ankle.

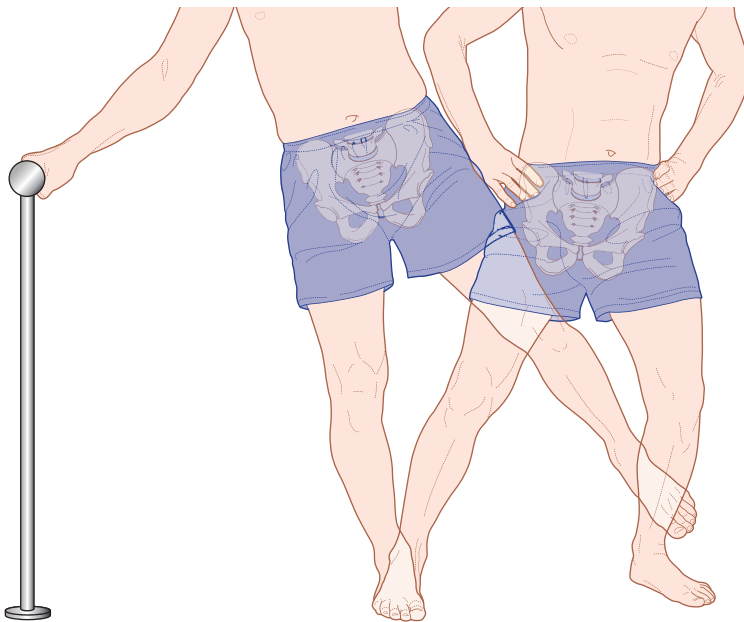


Figure 36.11 Hands on knee

To test myofascial resilience, put your hands on the medial and lateral side of an athlete’s knee (in this case, his right) to feel the “give” in the layers between your hands and the middle of the joint as he or she moves from side to side while maintaining the foot on the floor.

Perform this test on a dozen to a hundred people to clearly detect which tissues are deficient in either a resistant or lax condition.

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The knee is supposedly simply maintaining its extension between these two lateral movements at the joints above and below. But what do you feel inside and outside the knee in between as they carry out this movement? In a resilient body, you will feel a slight give in the tissues under your hand. In a healthy body, the skin will move the most, the underlying tissues of the anatomy trains (parietal myofasciae) will manifest some give, and, at the ligamentous level, a very slight give can be felt. With a little practice, you will be able to tell at which level there is too much give in the tissues or too little. With a little more practice you will be able to identify which layer within the leg is not coordinating with the rest, and apply to those tissues whatever treatment form is at your command.

Clinical summary

The new research on the properties of fascial tissues, the cybernetic nature of neural plasticity and the biotensegrity model of the actual engineering relationship between the soft and bony tissues, all suggest that holistic modeling of training results is required to supplement the individual structure model used for the last few centuries. The Anatomy Trains myofascial meridians map provides a model for measuring resilience and total body participation in movement that leads to long-term health. Learning to measure resilience and to recognize where tissue “give” is not occurring will allow the trainer to make adjustments, often at some distance from the site of stress or pain, which will promote maximum performance with minimum injury.

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